Nils Lukas

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Research Interests

Design secure and private AI agents in the presence of untrustworthy **providers**, **data**, **models**, and **users**.

1. **Providers**: Confidential computing & encrypted inference.
2. **Data**: Poisoning, prompt injection & training-data privacy.
3. **Models**: Scalable oversight & differentially private inference.
4. **Users**: Detect misuse via watermarking & safety monitoring.

Research Grants

Awarded **$1,458,000**

[United Al-Saqer Group] MedBoard: Privacy-preserving Multi-Agent AI for Healthcare **$136,000** · 2026
PI: **Nils Lukas** · Co-PIs: Praneeth Vepakomma, Preslav Nakov

[IAAI Research Program] Trust Layer for Safe Smallholder Advisory **$500,000** · 2026
PI: Salem Lahlou · Co-PIs: **Nils Lukas**, Martin Takáč

[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments **$100,000** · 2025
PI: **Nils Lukas** · Co-PI: Praneeth Vepakomma

[Etihad Airways] Conversational Booking Agents **$450,000** · 2025
PI: **Nils Lukas** · Co-PIs: Salem Lahlou, Alham Fikri Aji, Martin Takáč, Mingming Gong

[United Al-Saqer Group] Privacy-preserving Brain–Computer Interfaces **$136,000** · 2025
PI: Abdulrahman Mahmoud · Co-PIs: **Nils Lukas**, Elizabeth Churchill

[TII Funding] GFlowNets for Fuzzing of Agentic Applications **$136,000** · 2025
PIs: Salem Lahlou & **Nils Lukas**

Conference Publications

Underlined authors are students and postdocs I supervise as primary advisor.

25 **[EMNLP'26]** Main Conference AR: 15.4% (2,719/17,669)
CopyShield: A Cross-Level Benchmark of Copyright Defenses in LLMs
Maryam Alshehyari, Dushyant Chauhan, Samuele Poppi, Martin Takáč, Salem Lahlou, **Nils Lukas**. Conference on Empirical Methods in Natural Language Processing (Main Conference), 2026.

24 **[EMNLP'26]** Findings AR: 14.3% (2,533/17,669)
TRACER: Trace-Regularized Training Against Indirect Prompt Injection
Kshitij Mishra, Salem Lahlou, **Nils Lukas**. Findings of the Association for Computational Linguistics: EMNLP, 2026.

23 **[ECCV'26]**
SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation
Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, **Nils Lukas**, Karthik Nandakumar, Fakhri Karray, Samuele Poppi. European Conference on Computer Vision, 2026.

22 **[CVPR'26]** AR: 25.4% (4,090/16,092) 🏆 **Oral (top 0.8%)**
RAVEN: Erasing Invisible Watermarks via Novel View Synthesis
Fahad Shamshad, **Nils Lukas**, Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.

21
[ICML'26]
AR: 26.6%
(6,352/23,918)

Collaborative Threshold Watermarking
Tameem Bakr, Anish Ambreth, **Nils Lukas**. The Forty-Third International Conference on Machine Learning, 2026.

20
[ICML'26]
AR: 26.6%
(6,352/23,918)

CoRe: Collaborative Reasoning via Cross Teaching
Kshitij Mishra, Mirat Aubakirov, Martin Takáč, **Nils Lukas**, Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.

19
[ACL'26]
Main Conference
AR: 18.9%
(2,296/12,148)

Forest Before Trees: Latent Superposition for Efficient Visual Reasoning
Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, **Nils Lukas**, Yuhan Liu. The 64th Annual Meeting of the Association for Computational Linguistics, 2026.

18
[ACL'26]
Findings
AR: 17.8%
(2,163/12,148)

Detecting AI-Generated Video: A Vision-Language Dual-View Survey
Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, **Nils Lukas**, Gus Xia, Xiuying Chen, Yuhan Liu. Findings of the Association for Computational Linguistics, 2026.

17
[ICLR'26]
AR: 27.4%
(5,355/19,525)

DP-Fusion: Token-Level Differentially Private Inference for Large Language Models
Rushil Thareja, Preslav Nakov, Praneeth Vepakomma, **Nils Lukas**. The Fourteenth International Conference on Learning Representations, 2026.

16
[EACL'26]
Findings

SD-E²: Semantic Exploration for Reasoning Under Token Budgets
Kshitij Mishra, **Nils Lukas**, Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.

15
[NeurIPS'25]
AR: 24.5%
(5,290/21,575)

Mask Image Watermarking
Runyi Hu, Jie Zhang, Shiqian Zhao, **Nils Lukas**, Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.

14
[EMNLP'25]
AR: 22.2%
(1,811/8,172)

SPIRIT: Patching Speech Language Models against Jailbreak Attacks
Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, **Nils Lukas**. Empirical Methods in Natural Language Processing (Main Conference), 2025.

13
[ICML'25]
AR: 26.9%
(3,260/12,107)
★ Spotlight (Top 2.6%)

Optimizing Adaptive Attacks against Content Watermarks for Language Models
Abdulrahman Diaa, Toluwani Aremu, **Nils Lukas**. The Forty-Second International Conference on Machine Learning, 2025.

12
[ICML'25]
AR: 26.9%
(3,260/12,107)

Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems
Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, **Nils Lukas**, Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.

11
[USENIX'24]
AR: 19.1%
(417/2,176)

PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting
Rasoul Mahdavi, **Nils Lukas**, Faezeh Ebrahimianghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.

10
[USENIX'24]
AR: 19.1% (417/2,176)

Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimianghazani, Bailey Kacsmar, Xinda Li, **Nils Lukas**, Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.

9
[ICLR'24]
AR: 30.8% (2,250/7,262)

Leveraging Optimization for Adaptive Attacks on Image Watermarks
Nils Lukas, Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.

8
[ICLR'24]
AR: 30.8% (2,250/7,262)
◉ **Media Coverage**

Universal Backdoor Attacks
Benjamin Schneider, **Nils Lukas**, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.

7
[USENIX'23]
AR: 29.2% (422/1,444)

PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.

6
[S&P'23]
AR: 17.0% (195/1,147)
★ **Distinguished Contribution Award at Microsoft MLADS**

Analyzing Leakage of Personally Identifiable Information in Language Models
Nils Lukas, Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.

5
[S&P'22]
AR: 14.5% (147/1,012)

SoK: How Robust is Image Classification Deep Neural Network Watermarking?
Nils Lukas, Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.

4
[ICLR'21]
AR: 28.7% (860/2,997)
★ **Spotlight (Top 5%)**

Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
Nils Lukas, Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.

3
[IH&MMSEC'21]
AR: 40.3% (128/318)

On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
Masoumeh Shafieinejad, **Nils Lukas**, Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.

2
[ACSAC'20]
AR: 20.9% (104/497)

Practical Over-Threshold Multi-Party Private Set Intersection
Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, **Nils Lukas**, John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.

1
[EuroS&P'20]
AR: 20.9% (39/187)

Differentially Private Two-Party Set Operations
Bailey Kacsmar, Basit Khurram, **Nils Lukas**, Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.

PostDocs, Students

Postdoctoral Researchers

- Samuele Poppi — since 2025

- Kshitij Mishra *(secondary: Martin Takáč & Salem Lahlou)* since 2025
- Dushyant Chauhan *(secondary: Martin Takáč & Salem Lahlou)* since 2025
- Yuhan Liu *(secondary: Martin Takáč & Salem Lahlou)* since 2025

Research Engineers

- Sarim Hashmi since 2025

PhD Students

- Fabian Zhang (ML) since 2025
- Tair Djanibekov (NLP) since 2025
- Rushil Thareja (NLP) *(secondary: Praneeth Vepakomma)* since 2024
- Toluwani Aremu (ML) since 2023

MSc Students

- Juan Nicolas Sepulveda Arias (CV) since 2025
- Mohamed Mahmoud Mohamed Hendy (ML) since 2025
- Majid Mohamed Mohamed Sulaiman Ibrahim (ML) since 2025
- Prince Jha (ML) *(primary: Salem Lahlou)* since 2025
- Muhammad Mohideen (CV) *(secondary: Samuel Horvath)* since 2025

Alumni

- Tameem Bakr (ML) MSc · Mar 2026
- Shaikha Jasem Mohamed Ismail Alhosani (ML) MSc · Mar 2026
- Maryam Abdulla Rashed Abdulla Alshamsi (ML) MSc · Mar 2026
- Maryam Alshehyari (ML) MSc · Mar 2026
- Abdalla Khalid Mohamed Abdalla Alzaabi (ML) MSc · Mar 2026
- Ali Al-Ali (ML) MSc · Mar 2026

Talks & Keynotes

AI Safety Workshop: Alignment, Oversight and Deception

- ADIA Lab & UGR Summer School, Granada 2026

The Role of Watermarking for ML Security

- CyberQ, UAE Cyber Security Council & TII, Abu Dhabi 2025

Adaptively Robust and Forgery-Resistant Watermarking

- Meta (FAIR), hosted by Hady Elsahar 2025
- International Symposium on Trustworthy Foundation Models, MBZUAI 2025

Emerging Topics in Machine Learning

- Build with AI, Google Developer Groups, Abu Dhabi 2025

Optimizing Adaptive Attacks against Content Watermarks

- DeepMind, hosted by David Stutz 2024
- University of California, Berkeley, hosted by Dawn Song 2024

Analyzing Leakage of Personal Information in Language Models

- Microsoft M365, hosted by Robert Sim 2024
- Meta, hosted by Will Bullock 2023
- MongoDB, hosted by Marilyn George and Archita Agarwal 2023

How Reliable is Watermarking for Image Generators?

- Google, hosted by Somesh Jha 2023
- University of California, Berkeley, hosted by Dawn Song 2023

Keynotes

- **Aviation Future Week**, hosted by Emirates, Dubai 2024
- **Cyber Energy Leadership Forum**, Abu Dhabi 2024

Honors & Awards

Amazon Research Award **[100,000 USD]** 2025
First Place at the NeurIPS'24 Watermarking Competition **[4,400 USD]** 2024
First Place at DGE Elite Hackathon, GITEX'24 **[10,900 USD]** 2024
Top Mathematics Doctoral Thesis, University of Waterloo **[1,080 USD]** 2024

Alumni Gold Medal, One PhD Award Yearly, University of Waterloo 2024
Best Poster Award, Sponsored by David R. Cheriton **[220 USD]** 2023
Distinguished Contribution Award, Microsoft MLADS conference 2023
David R. Cheriton Scholarship, University of Waterloo **[14,400 USD]** 2022, 2023
Outstanding Reviewer, ICML'22 2022
Best Poster Award, Sponsored by Rogers **[720 USD]** 2019

Workshop Papers

3 **[GenAI4Health]** Sanitizing Medical Documents with Differential Privacy using Large Language Models
Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, **Nils Lukas**. 2025.

2 **[WMARK'25]** First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge
Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and **Nils Lukas**. The 1st Workshop on GenAI Watermarking, 2025.

1 **[WMARK'25]** ★ Oral Presentation
Optimizing Adaptive Attacks against Content Watermarks for Language Models
Abdulrahman Diaa, Toluwani Aremu, **Nils Lukas**. The 1st Workshop on GenAI Watermarking, 2025.

Working Papers

Mitigating Watermark Forgery in Generative Models via Randomized Key Selection, Submitted.
Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, **Nils Lukas**. 2025.

Robust and Calibrated Detection of Authentic Multimedia Content, Submitted.
Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, **Nils Lukas**. 2025.

Journal Publications

2 **[IEEE S&P'23]** Privacy-Preserving Machine Learning [Cryptography]
Florian Kerschbaum, **Nils Lukas**. IEEE Security & Privacy, Volume 21, Issue 6, 2023.

1 **[AIP'18]** SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization
Pascal Richter, Gregor Heiming, **Nils Lukas**, Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.

Service

Track Chair
- ACM Conference on Computer and Communications Security (CCS), Security and Privacy of Machine Learning (co-chair) 2027

Area Chair
- AAAI Conference on Artificial Intelligence (AAAI), Senior Program Committee 2027
- International Conference on Learning Representations (ICLR) 2026, 2027
- Neural Information Processing Systems (NeurIPS) 2026
- International Conference on Machine Learning (ICML) 2026

Program Committee
- ACM Conference on Computer and Communications Security (CCS) 2025
- IEEE Symposium on Security and Privacy (IEEE S&P) 2025–2027
- Recent Advances in Intrusion Detection (RAID) 2024

Artifact Evaluation Committee
- The ACM Conference on Computer and Communications Security (CCS) 2023, 2024

Reviewer

- European Conference on Computer Vision (ECCV) 2026
- Conference on Language Modeling (COLM) 2026
- British Machine Vision Conference (BMVC) 2026
- FAGEN Workshop @ ICML 2026
- Workshop on GenAI Watermarking (WMark @ ICLR) 2025
- NETYS 2025
- ACM TheWebConf (WWW) 2025
- International Conference on Learning Representations (ICLR) 2024, 2025
- International World Wide Web Conference (TheWebConf) 2024
- Recent Advances in Intrusion Detection (RAID) 2023
- Neural Information Processing Systems (NeurIPS) 2022, 2023
- International Conference on Machine Learning (ICML) 2022, 2025
- The Conference on Information and Knowledge Management (CIKM) 2020

Journal Reviewer

- Communications of the ACM (CACM) 2025

Other

- External Examiner, PhD Dissertation, CISPA Helmholtz Center for Information Security 2026
- Faculty Search Committee, Machine Learning Department at MBZUAI 2025–2026
- Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI 2025
- Admission's Committee, MBZUAI Machine Learning Department 2025
- Student Board Member, Cybersecurity and Privacy Institute 2022–2024
- Session Chair, IEEE Symposium on Security and Privacy (S&P) 2023
- Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS) 2021–2023
- School Advisory Committee on Appointments Liaison, CrySP Lab 2022
- Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA) 2018

Teaching

Instructor, MBZUAI, UAE

- ML8502: Machine Learning Security (14 weeks) 2025, 2026
- ML8509: Collaborative Machine Learning (w/ S. Horvath) 2026
- ML807: Federated Learning (7 weeks) 2025
- ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks) 2024

Teaching Assistant, University of Waterloo, Canada

- CS458/658: Computer Security and Privacy 2020, 2021
- CS246 – Object Oriented Programming 2021

Co-Instructor, RWTH-Aachen, Germany

- Course: Data-driven Medicine 2018

Education

University of Waterloo, Canada 2019 – 02/2024
Ph.D. in Computer Science

- Advisor: Florian Kerschbaum
- Thesis: Analyzing Threats of Large-Scale Machine Learning Systems
- Thesis Awards: Top Mathematics Doctoral Prize & Alumni Gold Medal

RWTH-Aachen, Germany
M.Sc. in Computer Science *(w/Distinction)* 2016 – 2018
B.Sc. in Computer Science 10/2012 – 2016

Work Experience

Assistant Professor, MBZUAI, Abu Dhabi, UAE since 08/2024
Research Intern, Royal Bank of Canada, Borealis AI, Toronto 2024

- Vertical Federated Learning, hosted by Kevin Wilson

Research Intern, Microsoft Research, Cambridge, UK 2022

- Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz

Research Assistant, RWTH-Aachen, Aachen 2014 – 2018
Student Researcher, DSA Daten- und Systemtechnik GmbH, Aachen 2016
Software Engineer Intern, A.R. Bayer DSP Systeme GmbH, Düsseldorf 2012